

ElTariffKatalog

The discovery layer of Eltariff-API

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The problem

- Sweden is introducing **time-differentiated power tariffs**
- The project's **original goal**: better support for **smart EV charging** and load shifting
- But there are **many grid operators**, each publishing **their own** tariffs
 - A deliberate **design choice** in this project

Given one facility, *where* do I fetch the right tariff data?

The ecosystem – three parts

Part	Role
Grid Tariff API	Each grid operator (DSO) publishes <i>its own</i> tariffs
Catalogue API	Central registry of <i>where</i> each operator's API lives
Clients	Consumers – apps, services, aggregators

The catalogue is the entry point. Clients call it *first*.

Used **primarily** during onboarding of a new user – not on every tariff fetch.

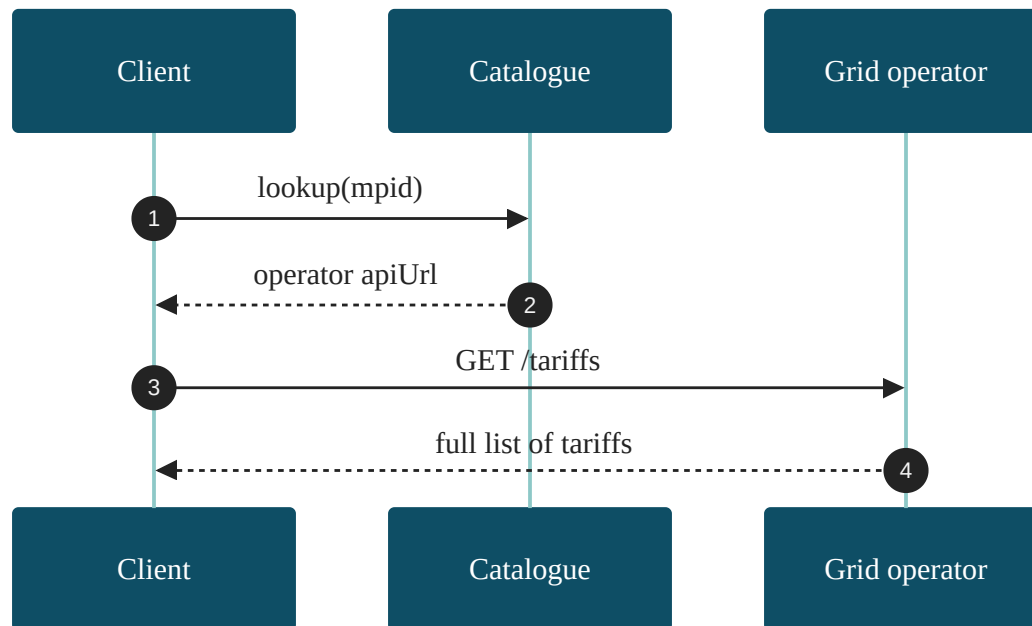
What the catalogue is

ElTariffKatalog — a central registry, i.e. a "phone book"

- Maps **18-digit metering-point ID ranges** → an operator's Grid Tariff API URL
- Holds **no tariff data itself** — only pointers
- Served at <https://eltariff.se>

How it works – the lookup flow

1. Client has a facility's metering-point ID (anläggnings-id)
2. Calls catalogue **lookup** → gets the operator's **apiUrl**
3. Calls that operator's **/tariffs** → gets the actual tariff data



Two sub-services

register-service

write · authenticated

Operators onboard and maintain their own entries.

supplier-service

read · public

Clients look up where to fetch tariffs.

- Register requires OAuth2 client-credentials (scope `register:write`, token at `id.deplide.org`) or a bearer JWT
- Lookup is public — no auth
- `409 CONFLICT` guards against overlapping MPID ranges

Endpoints

Method & path	Purpose
POST /tariffcatalogue/register	Register API info for an MPID range
PUT /tariffcatalogue/register	Update an entry
DELETE /tariffcatalogue/register/{supplierEntryId}	Remove an entry
POST /tariffcatalogue/lookup	Bulk lookup (many MPIDs)
GET /tariffcatalogue/lookup/{mpid}	Single lookup
GET /tariffcatalogue/all	Get the entire catalogue (all entries)

The data model

```
// SupplierEntry – one catalogue record
{
  "id": "uuid",
  "meteringPointIdFrom": "7300000000000000001",
  "meteringPointIdTo": "73000000000000000999",
  "companyName": "Example Grid AB",
  "companyOrgNo": "556000-0000",
  "apiUrl": "https://api.example.se/gridtariff",
  "userDocUrlOrEmail": "support@example.se"
}
```

EntryMapping: meteringPointId → entryId · MPID pattern `^[0-9]{18}$`

Status & timeline

- 1 Mar 2025 — Göteborg Energi & Tekniska verken first to publish
- Spring 2025 — RISE status website launches
- Goal: **all grid operators** publishing their tariffs in the API

Run by **RISE**, funded by **Energimyndigheten/FFI**. The **Tariff API** builds on a Norwegian solution; the **catalogue service is new** — Norway has only a static file of entries on GitHub.

Grid operators: Vattenfall, E.ON, Kraftringen, Halmstad, Jämtkraft and more.

Beyond DSOs: **Volvo Cars, Volvo Trucks, Einride, Tibber, Power Circle, Energiföretagen** (Swedenergy).

What's still evolving

Known caveats — API is in a **development phase** (feedback welcome). Conformance flags vs the Swedish API profile: verb-named resources, `entryMapping` should be plural, errors lack a body schema.

Under discussion / requested

- **Geographic lookup** — find catalogue entries by **position**, so digital service providers (e.g. for charging stations) need no MPID
- **Distributed catalogue** — integrate the catalogue API into the **Tariff API**, making the catalogue distributed too

Takeaways

The catalogue is the phone book of Eltariff-API:
look it up by facility ID → get the operator's API → fetch tariffs.

GitHub: <https://github.com/RI-SE/Eltariff-API>

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Thank you · Questions?